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Lab: Lab 3

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a. Program Description:

The purpose of this program is to calculate interest rate based on inputs given by the user and running these inputs in a while loop. The program shows how a while loop can be run multiple times to collect user inputs till conditions are met. Here we used input function to collect user inputs for principal, rate of interest, time and how many time a year the interest was compounded. Based on these values, total and interest paid was calculated. The user was asked to input the values multiple times, till the user entered 0 upon which the program exits. To print name and time,, function call was used and function defined in another program was imported into the source code of this program.

b. Program Source Code:

```
1  from print_me_first import *
2
3  print_me_time()
4
5  p = 1
6
7  while (p > 0):
8      try:
9          p = int(input("Enter the starting principle, 0 to quit: "))
10         if p == 0:
11             print("Program exiting ...")
12             exit()
13         r = float(input("Enter the annual interest rate: "))
14         n = int(input("How many times per year is the interest compounded? "))
15         t = float(input("For how many years will the account earn interest? "))
16         total = p * (1 + (r / (100 * n))) ** (n * t)
17         interest = total - p
18         print("At the end of {:.1f} years you will have $ {:.2f} with interest earned $ {:.2f} \n".format(t, total, interest))
19     except Exception:
20         print("Please enter a valid integer or float value only")
21
```

c. Program Output

```
C:\Users\shwetachauhan\PycharmProjects\pythonProject\venv\Scripts\python.exe C:/Users/shwetachauhan/PycharmProjects/Course_scripts/InterestCalculator.py
Name      : CNET-142 - Shweta Chauhan
Lab       : Lab 3 - Interest Rate
Current Time : Apr-03-2022 Sun (10:17:31PM)
Enter the starting principle, 0 to quit: 10000
Enter the annual interest rate: 1.00
How many times per year is the interest compounded? 400
For how many years will the account earn interest? 2
At the end of 2.0 years you will have $ 12,477.23 with interest earned $ 477.23

Enter the starting principle, 0 to quit: 10000
Enter the annual interest rate: 1.00
How many times per year is the interest compounded? 4
For how many years will the account earn interest? 1
At the end of 1.0 years you will have $ 3,056.70 with interest earned $ 56.70

Enter the starting principle, 0 to quit: 0
Program exiting ...

Process finished with exit code 0
```